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Energy-Enhanced Aqueous Organic Flow Batteries Enabled by Integrated Proton-Coupled Electron Transfer-Induced Carbon Capture and Utilization

Kaiping Zhu[1] | Sheng Xu[1] | Xingqi Wang[1] | Zhuwen Wei[1] | Kunlong Yang[1] | Zepeng Chen[1] | Xiaolin Wang[1] | Binglin Lu[1] | Xinglei Zhao[2] | Yan Jing[1]

[1]Department of Materials Science and Engineering, National University of Singapore, Singapore | [2]State Key Laboratory of Petroleum Pollution Control, CNPC Research Institute of Safety and Environmental Technology, Beijing, China

Correspondence: Yan Jing (yan-jing@nus.edu.sg)

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ABSTRACT

Aqueous organic redox flow batteries offer a safe, scalable, and potentially cost-effective solution for storing massive electricity from intermittent renewables. Meanwhile, operating at ambient temperature and pressure, electrochemically induced carbon capture using aqueous redox sorbents in flow systems has recently emerged as a low-energy technology for climate change mitigation. Integrating these two technologies within an aqueous organic flow system represents a unified route to achieving decarbonization. Here, employing organic molecules undergoing proton-coupled electron transfer as redox species, we establish aqueous all-organic flow battery systems that achieve enhanced energy storage as well as coupled CO_2 capture and utilization. During discharge, CO_2 from simulated flue gas is captured as HCO_3^- by the oxygen-tolerant positive electrolyte (posolyte). The buffered posolyte prevents negative shifts in its redox potential, elevating a flow cell voltage up to 1.32 V, and enhancing discharge energy by at most 24.3% compared to operations under N_2. During charge, CO_2 is released with a CO_2/e molar ratio approaching unity at an energy cost of 30.65 kJ mol^{-1} CO_2. As a proof-of-concept, our work establishes a new paradigm for integrating enhanced energy storage with carbon capture, utilization, and storage, paving the way for next-generation multifunctional electrochemical redox flow systems.

1 | Introduction

To mitigate climate change, it is essential to not only accelerate the adoption of renewable energy but also capture CO_2 from hard-to-abate sectors. Composed of earth-abundant elements, redox molecules-based aqueous organic flow batteries show promise in storing large-scale electricity from intermittent renewables with greatly reduced levelized costs of electricity storage [1–5]. Powered with renewable electricity, electrochemically induced carbon capture using organic redox sorbents in aqueous flow systems have shown potential in significantly lowering the energy consumption as it operates at ambient temperature and pressure [6–15]. Given the common reliance on redox molecules and aqueous flow electrochemistry in the two systems, it is reasonable to integrate both functions into a single system when appropriate, as it eliminates the need for two separate flow systems, thereby significantly reducing capital expenditure on equipment and materials [13, 16].

Owing to the great reversibility, facile kinetics, and structural diversity, organic molecules undergoing proton-coupled electron transfer (PCET) such as anthraquinones [1, 17, 18], phenazines

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[19–21], azobenzene [22, 23], azopyridinium [24], fluorenones [25, 26], have been extensively studied as redox species in aqueous flow batteries across the full pH range. When a PCET molecule-based electrolyte operates between mildly acidic and strongly alkaline pH, it can induce a pH-swing that enables reversible CO_2 capture and release. Theoretically, a PCET molecule can hold dual functionality of serving as redox species in an aqueous organic flow battery electrolyte and a redox sorbent in electrochemically induced carbon capture system [7, 9, 10, 13].

Despite the promise of integrating energy storage and carbon capture in aqueous organic flow battery systems with PCET redox molecules, existing studies are scarce and often suffer from either low energy density or poor CO_2 capture capacity in the presence of O_2 [9, 13, 14, 16, 27]. The challenge is largely attributed to the limited availability of water-soluble, oxygen-tolerant PCET molecules operating at near-neutral and alkaline pH [10, 28–30]. Most reported PCET molecules that operate within the desired pH range often exhibit high oxygen sensitivity, resulting in irreversible CO_2 capture and release [14, 31]. While a few PCET molecules show great oxygen tolerance and reversibility only under acidic conditions, they tend to become irreversible or structurally unstable at near-neutral or alkaline pH [24, 32, 33].

Here, utilizing phenazine derivatives [19, 30] as the redox species in negative electrolyte (negolyte) and oxygen-tolerant fluoflavine disulfonate (FFDS) [29] as the redox species in positive electrolyte (posolyte), we developed all-organic aqueous flow battery systems that exhibit enhanced energy storage when coupled with CO_2 capture. Compared to the operation in N_2, when a stream of simulated flue gas comprising 10% CO_2, 10% O_2, and 80% N_2 flows through FFDS posolyte, the flow battery shows an elevated open circuit voltage up to 1.32 V, increased discharge energy by 24.3%, and a round-trip energy efficiency of 73.9%. The oxygen-tolerant FFDS sorbent enables reversible CO_2 capture and release from the simulated flue gas, affording a CO_2/e molar ratio approaching unity with an energy cost of 30.65 kJ mol^{-1} CO_2 at 20 mA cm^{-2}. Our findings suggest that PCET molecule-based aqueous organic flow batteries provide a viable platform for coupling energy storage and carbon capture. Notably, when an oxygen-tolerant PCET molecule-based posolyte is employed for carbon capture, the flow battery exhibits an increased cell voltage and enhanced overall energy output, highlighting a synergistic benefit of integration.

2 | Result and Discussion

Figure 1 depicts the simultaneous voltage enhancement and CO_2 capture, utilization and storage (CCUS) in an aqueous organic flow battery system undergoing PCET. During discharge, by directly feeding flue gas into the water-soluble, oxygen-tolerant, high-potential FFDS posolyte, the posolyte pH will be dynamically buffered through the reversible CO_2/HCO_3^- equilibrium, preventing the redox potential of FFDS from drifting negatively, affording an elevated overall discharge voltage and enhanced energy density, relative to the operation in the absence of CO_2 in posolyte. During charge, the captured CO_2 is released from the electrolyte for storage. The oxygen-tolerance of FFDS enables reversible CO_2 capture and release, preventing the accumulation of dissolved inorganic carbons in the electrolyte.

To demonstrate the proof of concept, we employed 2,2'-(phenazine-2,3-diylbis(oxy))dipropionate (D_2PEPz) and FFDS (Figure 2a) as the PCET redox molecules with the configuration of (–) 0.1 M D_2PEPz | 0.1 M FFDS (+) for flow battery testing as well as electrochemical CO_2 capture [30]. Detailed schematic experimental setup is shown in Figure S1. As shown in Figure 2b, the redox potential of D_2PEPz negolyte shifts positively as the pH decreases, its PCET process is further validated by the Pourbaix diagram in Figure 2c. Despite the PCET behavior, using D_2PEPz negolyte for electrochemically induced CO_2 capture presents two notable drawbacks [9, 13, 14]. First, the CO_2 capture-induced positive shift in the redox potential of D_2PEPz leads to a decrease in the open-circuit voltage of the flow battery. Second, because the redox potential of D_2PEPz is significantly lower than that of the oxygen reduction reaction (ORR), the reduced D_2PEPz is oxygen-sensitive and incapable of capturing CO_2 from O_2-containing flue gas. The cyclic voltammograms and the Pourbaix diagram of FFDS (Figure 2c) indicate it undergoes PCET process throughout the entire pH range. Furthermore, the redox potential of FFDS exceeds the potential of the two-electron transfer ORR [29], rendering it oxygen-tolerant for practical CO_2 capture. Moreover, when CO_2 is introduced to the FFDS posolyte during electrochemical redox reactions, the CO_2/HCO_3^- equilibrium creates a buffering effect that prevents the electrolyte pH from rising excessively, thus elevating the redox potential of FFDS. Rather than lowering the cell voltage, the positive shift in FFDS redox potential would substantially enhance it. It is noteworthy that during the reduction process, CO_2 does not undergo reduction due to thermodynamic constraints and the absence of a catalyst.

To experimentally validate our analysis, we investigated a flow cell with the configuration of (–) D_2PEPz | FFDS (+) under different gas environments by introducing CO_2 separately to the electrolytes (Figure 2d). Compared to the flow battery operating in N_2, the midpoint discharge voltage decreases by 0.21 V when CO_2 is introduced to D_2PEPz negolyte (Figure S2a); whereas it increases by 0.21 V when CO_2 is introduced to FFDS posolyte (Figure S2d), corroborating our analysis and the potential shifts observed in Figure 2b. Introducing CO_2 to D_2PEPz decreases the discharge energy output by 37.6%, whereas introducing CO_2 to FFDS increases it by 24.3% (Figure S2b,c and Table S1). In situ pH monitoring reveals that the capacity-limiting electrolyte pH fluctuates between 6.2 and 13.3 in N_2, whereas the presence of CO_2 constrained the electrolyte pH between 6.4 and 11 (Figure 2e; Figures S3 and S4). With the measured redox potentials and the pH values in the presence and absence of CO_2, we simulated the voltage profiles using Nernst equation (Figure 2f). The simulated voltage profile further confirms the enhancement in discharge voltage and energy output when CO_2 is introduced to FFDS (Figure S2e,f). Both the experimental and simulated results emphasize that an oxygen-tolerant posolyte undergoing PCET is crucial for achieving both reversible CO_2 capture and enhanced energy density.

To determine whether the elevated cell voltage could be maintained, we performed 100 consecutive charge–discharge cycles while continuously flowing simulated flue gas (80% N_2, 10% O_2, 10% CO_2) through the FFDS posolyte (Figure 3a–c). The constrained pH swing suggests that the posolyte is effectively buffered by the dynamic CO_2/HCO_3^- equilibrium, ensuring

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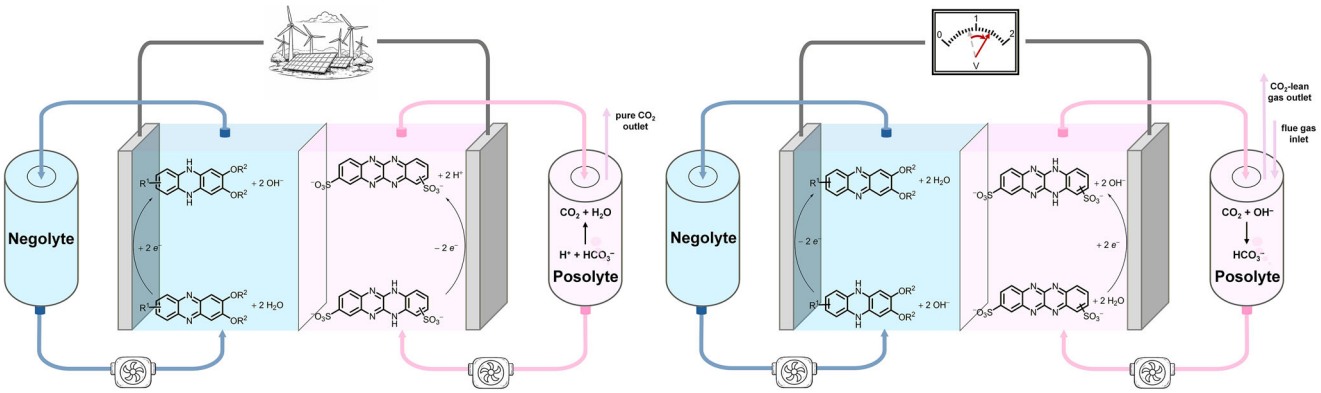

FIGURE 1 | Schematics of enhanced energy storage coupled with CCUS in a flow battery system. a, Charge coupled with CO_2 release for storage. During the charge, electrochemical oxidation acidifies the posolyte, releasing the captured CO_2 for storage. b, Discharge coupled with carbon capture, utilization (CCU) for enhanced cell voltage output. During the discharge, electrochemical reduction of the posolyte raises the electrolyte pH, facilitating CO_2 absorption from flue gas. Compared to the flow battery operating in N_2, the flow battery coupled with CO_2 capture in posolyte exhibits an enhanced cell voltage, as a result of buffering effect induced by the captured CO_2 in the posolyte.

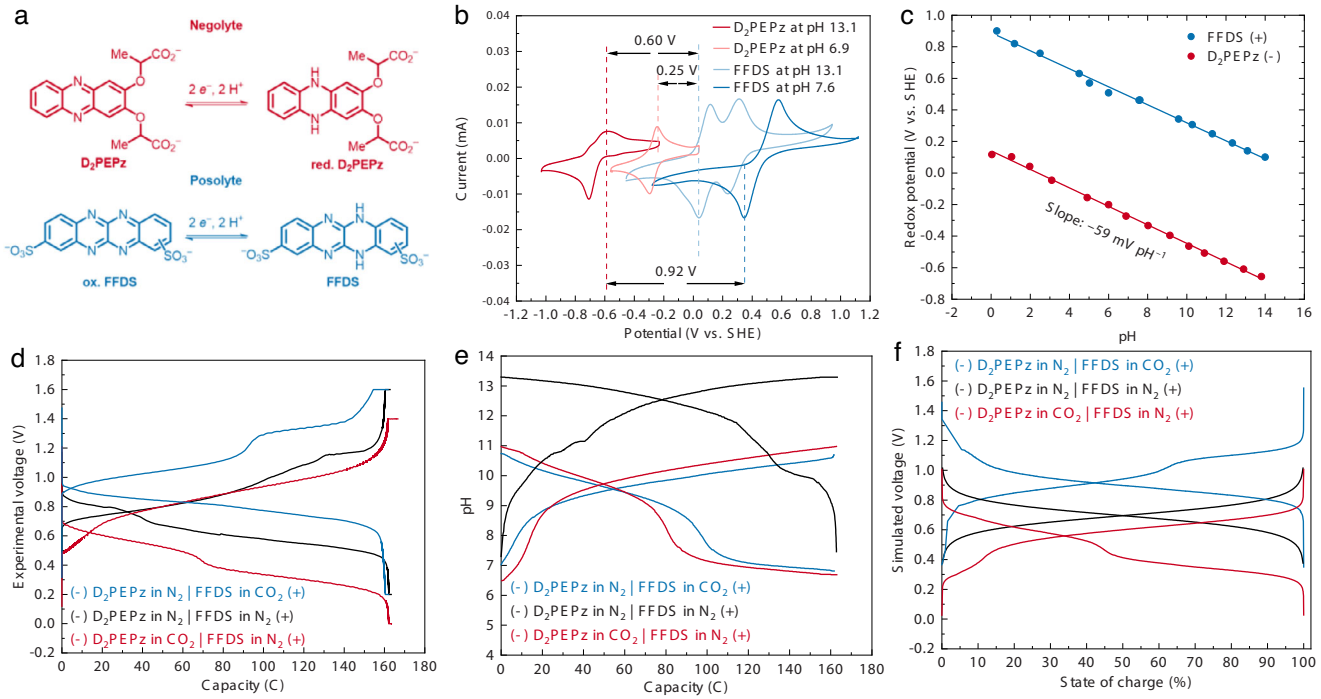

FIGURE 2 | Electrochemical characterizations of the flow cell with the configuration of (−) D_2PEPz | FFDS (+) under different gas environments. a, Chemical structure of D_2PEPz negolyte and FFDS posolyte. b, Cyclic voltammograms of D_2PEPz at pH 6.9 (buffered), 13.1 and FFDS at pH 7.6 (buffered), 13.1. c, Pourbaix diagram of D_2PEPz and FFDS. d, Experimental charge–discharge voltage profiles. e, Posolyte pH variation during charge–discharge process in N_2 (black) and CO_2 (blue) respectively, negolyte pH variation in CO_2 (red). f, Simulated voltage profiles based on the measured pH variation in Figure 2e. d–f, Simulated flue gas composed of 10% CO_2, 10% O_2, and 80% N_2 was used as the feed gas for FFDS posolyte. Simulated flue gas composed of 10% CO_2, and 90% N_2 was used as the feed gas for D_2PEPz negolyte.

long-term stability of voltage enhancement. The constant negolyte pH throughout cycling indicates that the CO_2 and HCO_3^- species exclusively buffer the posolyte and do not migrate toward the negolyte. Otherwise, the negolyte pH would decrease, causing its potential to rise and counteracting the voltage gain induced by the posolyte. In contrast, under an N_2 atmosphere, the pH of FFDS swings between 8.5 and 13.3, leading to a lower discharge potential (Figure 2d; Figure S5). The stable voltage profiles across different cycles (Figure 3d) and current densities (Figure 3e) confirm the posolyte-induced voltage gain over the prolonged operations. The factors affecting polarization observed in the rate performance (Figure 3e) include the pH difference induced by CO_2 capture and release processes, Ohmic resistance, mass transfer resistance, and charge transfer resistance [34]. This flow battery exhibits good cycling stability, >99% Coulombic efficiency, and high rate capability across 10–60 mA cm^{-2},

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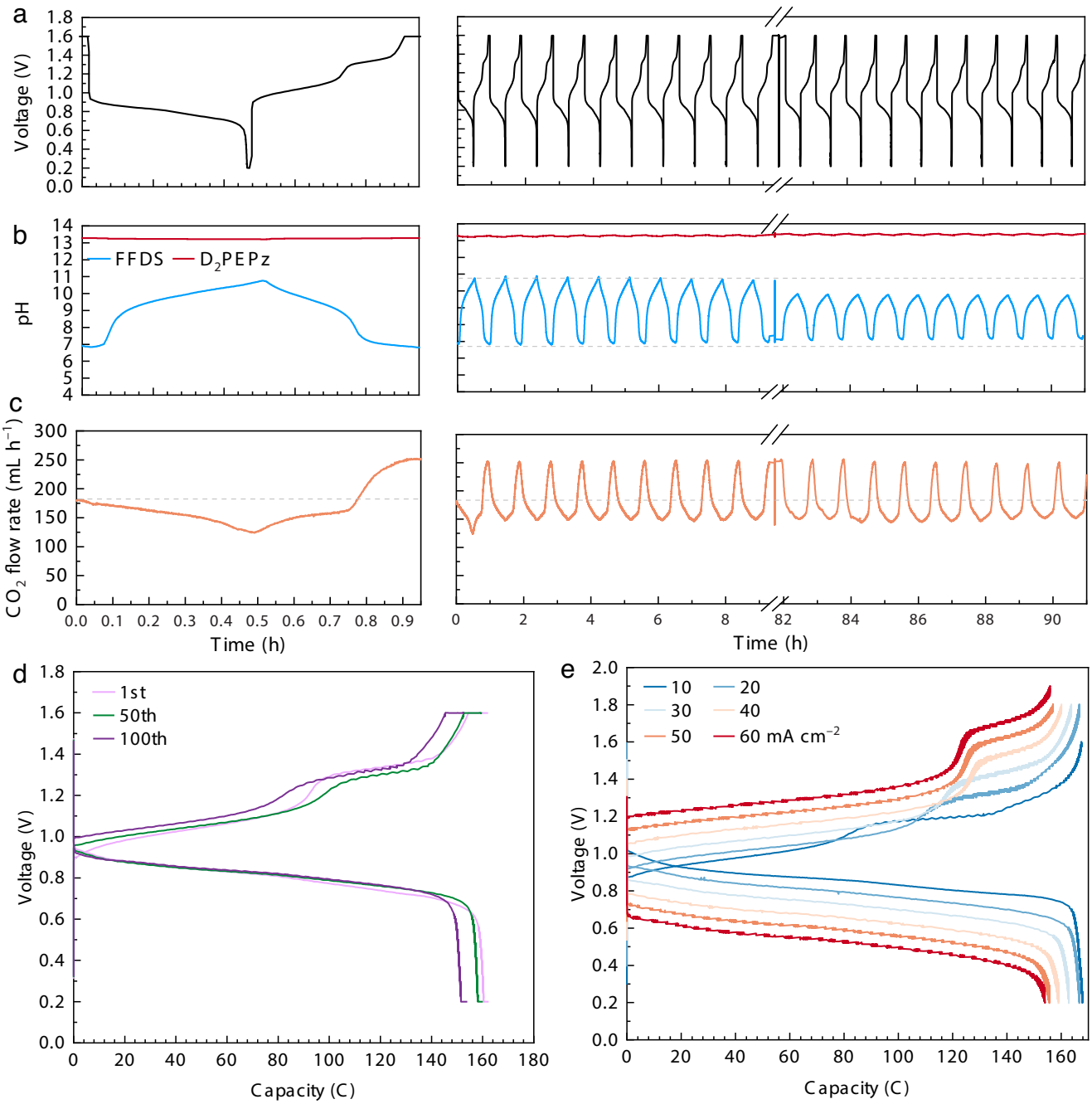

FIGURE 3 | Cycling of the flow cell with the configuration of (–) D_2PEPz in N_2 | FFDS in CO_2 (+). A stream of simulated flue gas composed of 80% N_2, 10% O_2, 10% CO_2 is introduced to FFDS posolyte at a flow rate of 30 mL min^{-1}. The D_2PEPz negolyte is under N_2. a, Voltage profile, b, pH variation, c, CO_2 flow rate, d, charge–discharge curves at different cycles, and e, charge–discharge curves under different current densities of 10 mL 0.1 M FFDS, 0.4 M KOH, 0.6 M KCl under simulated flue gas (80% N_2, 10% O_2, 10% CO_2) against 30 mL 0.2 M D_2PEPz, 0.4 M KOH, 0.6 M KCl (50% SOC) under 100% N_2.

highlighting the adaptability of the system in the presence of CO_2 (Figures S6 and S7). Impedance measurements show similar ohmic resistance under N_2 and simulated flue gas, implying that the voltage difference arises from CO_2-mediated pH regulation rather than resistance changes (Figure S8). Additionally, the incorporation of dissolved inorganic carbon exerts negligible influence on mass transport and charge-transfer kinetics of the PCET molecules (Figures S9 and S10). The flow battery delivers a peak power density of 68.56 mW cm^{-2}, with membrane resistance identified as the primary limitation to power output (Figure S11).

Ideally, CO_2 could function simultaneously as a buffer to enhance voltage and as the target for carbon capture. The stable oscillation in CO_2 flow rate suggests a reversible CO_2 capture and release (Figure 3c). However, the limited CO_2 capture capacity and CO_2/e molar ratio result from the lack of rest intervals between charge and discharge.

Rest intervals are introduced between charge and discharge to maximize CO_2 capture capacity. Figure 4a presents the voltage profiles with and without rest. The rest after charge facilitates

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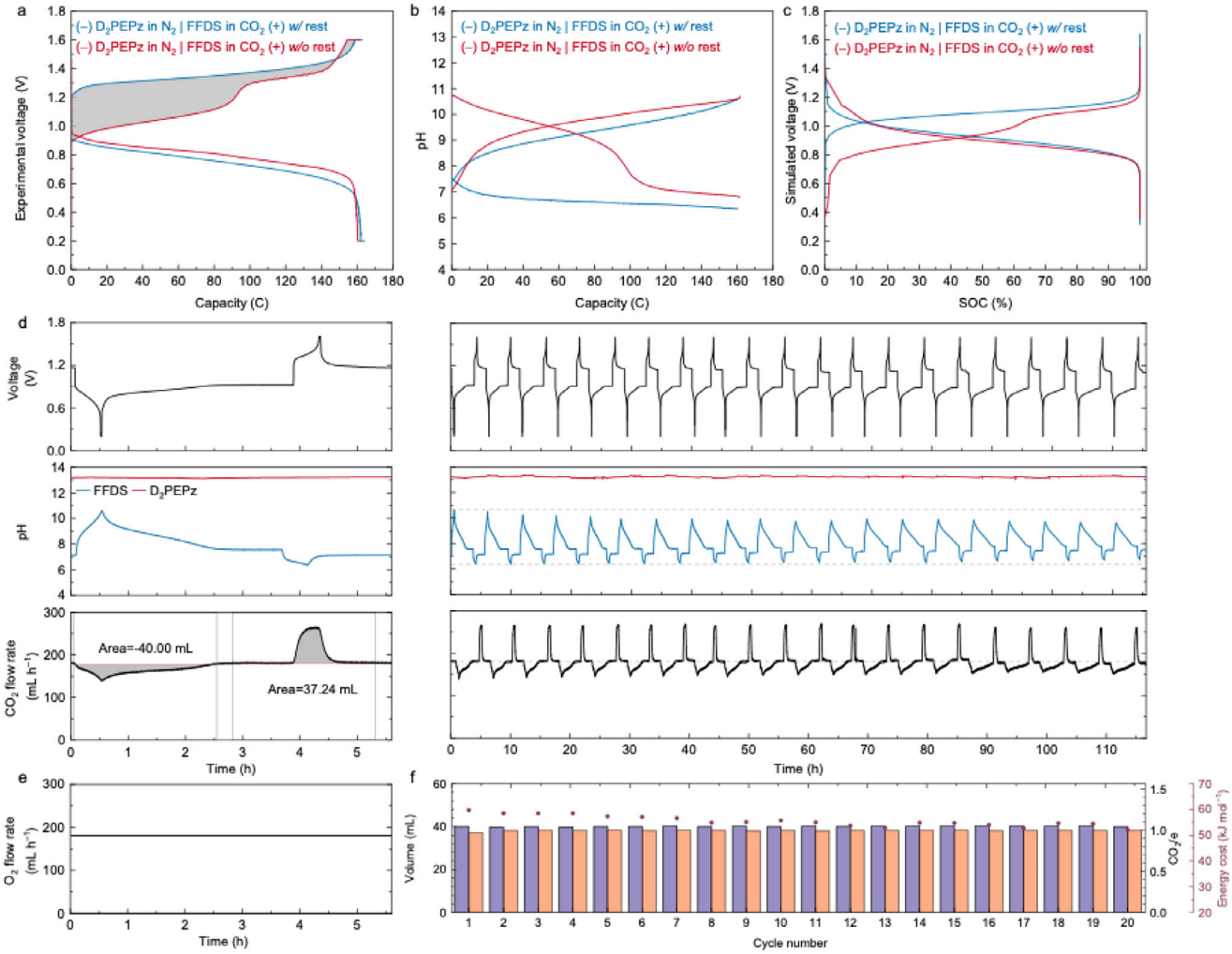

FIGURE 4 | Electrochemically induced CO_2 capture and release performance of the (–) D_2PEPz in N_2 | FFDS in CO_2 (+). a–c, Comparison of experimental voltage profiles, pH variations, and simulated voltage profiles of (–) D_2PEPz in N_2 | FFDS in CO_2 (+) with and without rest after charge and discharge processes. d, Voltage profile, pH variation, and CO_2 flow rate. e, O_2 flow rate. f, CO_2 volume profiles of 10 mL 0.1 M FFDS, 0.4 M KOH, 0.6 M KCl under simulated flue gas against 30 mL 0.2 M D_2PEPz, 0.4 M KOH, 0.6 M KCl (50% SOC) under 100% N_2.

full CO_2 release, without influencing the subsequent discharge voltage (Figure S12a). The rest after discharge enables the FFDS posolyte to complete CO_2 capture, as reflected by the continued pH decrease of posolyte from 10.9 to 7.5 (Figure 4b and Figure S12b). This pH reduction leads to an increase in the following charge voltage, lowering the round-trip energy efficiency (Figure 4a and Figure S12a). The decrease in energy efficiency arises from the energy consumption associated with the carbon capture, which is also confirmed by the simulated voltage profiles (Figure 4c). Figure 4d presents the variation of voltage, pH, and CO_2 flow rate over 20 cycles. The flow cell exhibits remarkably consistent and reversible voltage profiles (Figure S13), with no significant drift or polarization, suggesting excellent stability of the system. The FFDS posolyte pH initially fluctuates between 7.11–10.7 during the capture–release process and gradually converged to a narrower range of 6.69–9.67 as cycling proceeds, indicating that as CO_2 is completely absorbed, the formation of HCO_3^- species in the posolyte can effectively buffer the pH fluctuations. As the FFDS electrolyte is oxygen-tolerant, the O_2 flow rate maintains constant over time (Figure 4e). Throughout the 20 continuous CO_2 capture–release cycles (Figure 4f), FFDS posolyte exhibits an average CO_2 uptake of 37.89 mL per cycle (Figure S14), corresponding to a CO_2/e molar ratio close to **1** and an average energy cost of 55.54 kJ mol^{-1} CO_2 (Table S2). Our simulation indicates that the minimum energy cost is 16.06 kJ mol^{-1} CO_2 (Table S2), suggesting that further optimization could yield significant reductions in energy consumption. By introducing simulated flue gas into the oxygen-tolerant FFDS posolyte, we demonstrate a synergistic coupling of CO_2 capture and enhanced energy storage in a flow battery, highlighting the potential of this system for scalable, low-energy carbon capture alongside improved energy storage.

To further demonstrate the universality of the CO_2 dynamic regulation strategy for voltage enhancement, we employed an alternative phenazine negolyte i.e. 7,8-dihydroxyphenazine-2-sulfonic acid (DHPS) and constructed a high-voltage redox flow battery system [19]. As shown in Figure 5a, the DHPS negolyte exhibits a low redox potential of –0.89 V at pH 14. In situ pH measurements reveal that the pH of FFDS posolyte fluctuates within a narrower range in the presence of CO_2 (Figure 5b), while the DHPS negolyte in N_2 remains consistently stable at

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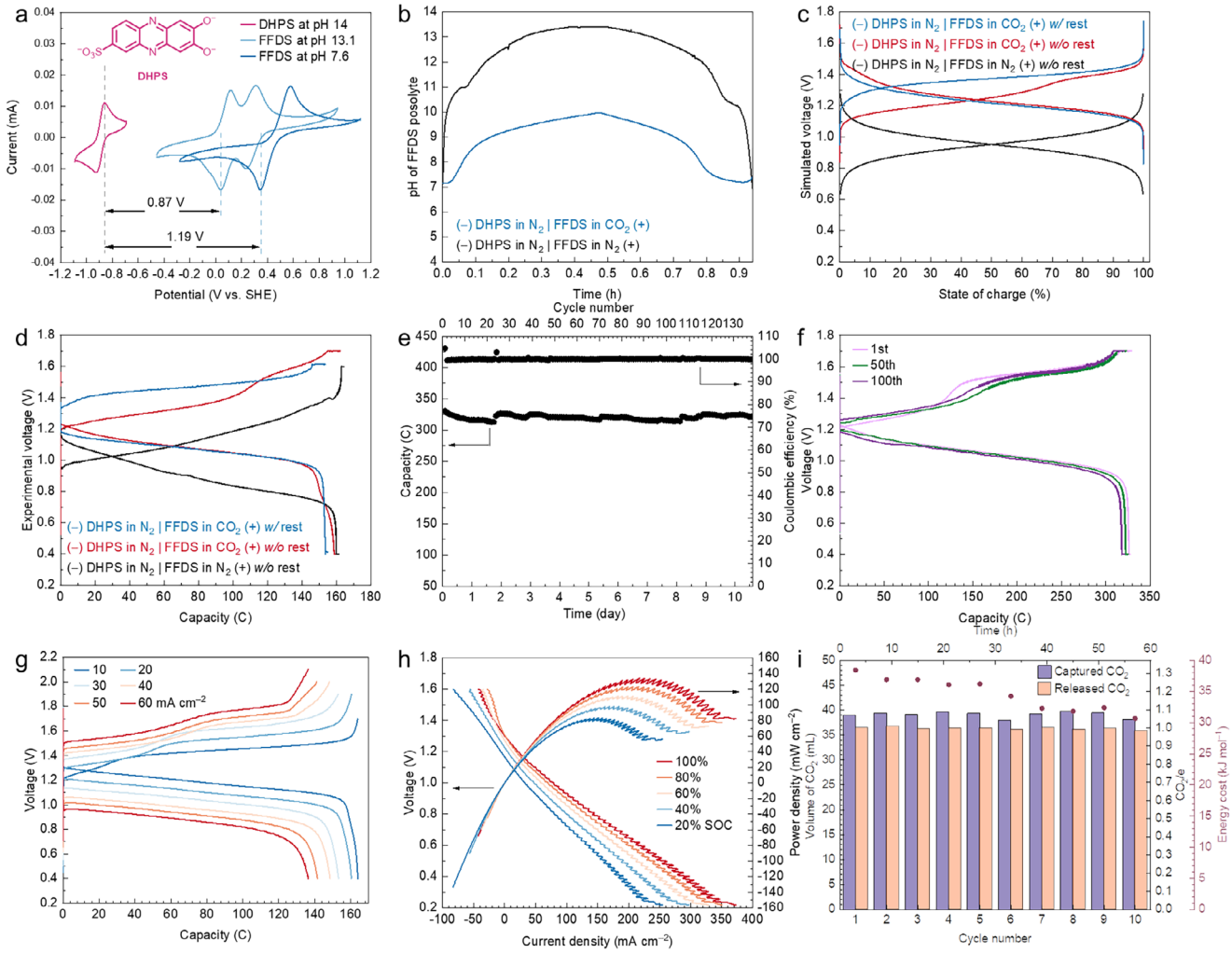

FIGURE 5 | Electrochemical characterizations of the flow cell with the configuration of (–) DHPS | FFDS (+) under different gas environments. a, Cyclic voltammograms of DHPS (at pH 14) and FFDS (at pH 13.1, 7.6). b, pH variation, c, experimental charge–discharge profiles, d, simulated charge–discharge profiles. e, Long-term cyclic performance, and f, charge–discharge profiles at different cycles of 10 mL 0.2 M FFDS, 0.8 M KOH, 0.6 M KCl under simulated flue gas (80% N_2, 10% O_2, 10% CO_2) against 30 mL 0.2 M DHPS (50% SOC), 1.6 M KOH under 100% N_2. g, Charge–discharge curves under different current densities, h, cell voltage and power density during discharge, and i, CO_2 capture–release cycling performance with rest intervals after charge and discharge steps. c, d, g–i: 10 mL 0.1 M FFDS, 0.4 M KOH, 1 M KCl under 100% N_2 and simulated flue gas (80% N_2, 10% O_2, 10% CO_2) against 30 mL 0.2 M DHPS (50% SOC), 1.6 M KOH under 100% N_2.

≈14 (Figure S15), suggesting that the migration of CO_2/HCO_3^- to negolyte is negligible. In contrast, the pH of FFDS in N_2 swings between 7.8 and 13.3, resulting in a lower discharge voltage (Figure S16, Figure 5d). The simulated voltage profiles in Figure 5c show a 0.22 V increase in discharge voltage and a 29.02% enhancement in energy (Figure S17a and Table S3). Such improvements have also been confirmed by our experimental results (Figure 5d), which exhibit a 0.20 V increase in discharge voltage and a 17.21% rise in energy output (Figure S17b and Table S3). Rest intervals enhance CO_2 capture by enabling full release after charge and continued absorption after discharge (Figure S18), which raises the subsequent charge voltage (Figure 5d) and slightly reduces energy efficiency due to the energy cost of carbon capture, consistent with simulation results (Figure 5c). The introduction of CO_2 exerts negligible influence on electrolyte resistance (Figure S19), validating that the discrepancy in cell voltage stems from the CO_2-indcued electrolyte pH variation. The flow battery exhibits a temporal fade rate of 0.59%/day and a cycle-denominated fade rate of 0.05%/cycle (Figure 5e). Compared to operation in N_2 (Figure S21), exposure of the posolyte to CO_2 results in consistently elevated voltage profiles across multiple cycles (Figure 5f) and current densities (Figure 5g, Figure S20), confirming the sustained posolyte-induced voltage gain during prolonged operation. The polarization in Figure 5h shows that the peak power density reaches up to 134.3 mW cm^{-2} when CO_2 is introduced to the posolyte, which is three times of the value delivered by the flow cell operating under N_2 conditions (Figure S22). It should be emphasized that a reduction in membrane resistance would lead to a further improvement in peak power density (Figure S23). Serving as the sorbent, FFDS delivers theoretical CO_2 capture capacity with a CO_2/e molar ratio approaching 1 and at a low energy cost of 30.65 kJ mol^{-1} CO_2 (Figure 5i and Figures S24 and S25). Capturing from 10% CO_2, 10% O_2, and 80% N_2, then releasing to 100% CO_2 with our system results in a total cost range of 36.55–44.34 kJ mol^{-1}. Optimizations in cell components and electrolyte composition

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would further reduce the resistance, approaching the theoretical energy cost of 19.69 kJ mol^{-1} (Table S4). Overall, the CO_2-based dynamic regulation strategy for voltage management is generally applicable to aqueous-soluble molecules that undergo PCET processes. The most pronounced voltage enhancement upon CO_2 introduction is expected for candidate molecules that combine high redox potential, high aqueous solubility, and high pK_a values in their reduced forms.

3 | Conclusions

In conclusion, by introducing simulated flue gas (10% CO_2, 10% O_2, 80% N_2) into an oxygen-tolerant PCET-enabled FFDS posolyte while maintaining the phenazine negolyte under N_2, we establish an aqueous all-organic flow battery that achieves reversible CO_2 capture and release at low energy cost, alongside elevated discharge voltage and enhanced energy density. The PCET-induced pH swing governs a reversible CO_2/HCO_3^- equilibrium in the FFDS posolyte, providing dynamic buffering and stabilizing the redox potential of FFDS against negative shifts. Both experimental and theoretical analyses confirm the effectiveness of this strategy, with flow batteries delivering an open-circuit voltage up to 1.32 V, a 24.3% enhancement in energy output under simulated flue gas, and a near-unity CO_2/e molar ratio at an energy cost of 30.65 kJ mol^{-1} CO_2. These findings highlight a viable pathway to synergistically integrate energy storage and carbon capture within a single flow system. Looking ahead, the rational design of oxygen-tolerant, water-soluble PCET molecules, together with advances in flow cell engineering, will offer ample opportunities to advance scalable, and cost-effective flow battery systems for enhanced energy storage and carbon capture, utilization, and storage.

4 | Methods

4.1 | Electrolytes Preparation

The posolyte is composed of 10 mL 0.1 M FFDS, 0.4 M KOH and 0.6 M KCl. After oxidation of the posolyte, the pH is adjusted to 7.

The negolyte is composed of 30 mL 0.2 M D_2PEPz, 0.4 M KOH and 0.6 M KCl (initial pH is adjusted to 7) or 30 mL 0.2 M DHPS and 1.6 M KOH. A flow cell was assembled by pairing the negolyte with potassium ferrocyanide, and the negolyte is reduced to 50% SOC.

4.2 | Electrochemical Characterizations

4.2.1 | Assembly of Flow Battery in the Presence and Absence of CO_2

For D_2PEPz capacity limitation side, 10 mL 0.1 M D_2PEPz, 0.2 M KOH, 0.8 M KCl under simulated flue gas (90% N_2, 10% CO_2) or 100% N_2 against 30 mL 0.1 M FFDS, 0.4 M KOH, 0.6 M KCl (50% SOC) under 100% N_2.

For FFDS capacity limitation side, 10 mL 0.1 M FFDS, 0.4 M KOH, 0.6 M KCl under simulated flue gas (80% N_2, 10% O_2, 10% CO_2) or 100 N_2 against 30 mL 0.2 M D_2PEPz, 0.4 M KOH, 0.6 M KCl (50% SOC) under 100% N_2.

For high-voltage flow battery, 10 mL 0.1 M FFDS, 0.4 M KOH, 1 M KCl under simulated flue gas (80% N_2, 10% O_2, 10% CO_2) or 100 N_2 against 30 mL 0.2 M DHPS, 1.6 M KOH (50% SOC) under 100% N_2.

The membrane is Nafion 117 or 212 cation exchange membrane. The electrodes are 10 pieces of baked WOS1009 carbon cloths on each side. The current density is 20 mA cm^{-2} followed with potential holds. The feed gas flow rate is 30 mL min^{-1}.

4.2.2 | Gas Sensor for Capture and Release

CO_2 data was recorded with 100% CO_2 Sensor (SPRINTIR-WX CM-0221). The downstream CO_2 reading (ppm) was monitored with CO_2 sensors and the GasLab data logging software. O_2 data was recorded by 25% Oxygen Flow Through Sensor (LOX-O_2-F UV Flux CM-42991).

4.2.3 | CO_2-Enhanced Energy Storage

During charge, the flow cell is charged at 20 mA cm^{-2} with potential holds at 1.6 V until the current density drops below 2 mA cm^{-2}. During discharge, the flow cell is discharged at 20 mA cm^{-2} with potential holds at 0.2 or 0.4 V until the current density drops below 2 mA cm^{-2}. A pH meter and a CO_2 sensor are used to monitor the pH change and CO_2 flow rate, respectively. 80% N_2, 10% O_2 and 10% CO_2 as the feed gas is continuously introduced to the posolyte at a flow rate of 30 mL min^{-1}. The negolyte is always maintained in a N_2 atmosphere.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All experimental data are available in the main text or Supporting Information. Source data are provided in this paper.

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